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DC Power Flow in Power Markets

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Executive Summary

DC Power Flow and DC Optimal Power Flow in Electricity Markets

Electricity markets operate within the physical constraints of the transmission grid. While market clearing determines which generators produce electricity and at what price, the resulting dispatch must also satisfy the physical laws governing power flows. These include power balance at each node (Kirchhoff's current law), loop flow behavior (Kirchhoff's voltage law), and thermal limits on transmission lines. If these constraints are not properly represented, a market outcome may appear economically optimal but be physically infeasible, requiring costly redispatch by the system operator.

The most complete representation of the grid is given by the AC load flow model. This model describes voltage magnitudes, voltage angles, active power, and reactive power through nonlinear equations. Although accurate, AC load flow is computationally demanding and not well suited for repeated optimization, which is required in market simulations and market-clearing algorithms. In electricity markets, we are primarily concerned with active power, production costs, congestion, and prices. Under typical transmission system conditions, where voltage magnitudes are close to 1 per unit, transmission line resistances are small relative to reactances, and voltage angle differences are small, the AC equations can be significantly simplified. These simplifications lead to the DC load flow model.

The DC load flow provides a linear relationship between active power injections and voltage angles. In matrix form, injections are proportional to voltage angles through a susceptance matrix. Line flows can then be expressed as linear functions of angle differences. This linear structure is crucial: it allows the physical grid model to be embedded directly into linear optimization problems. Unlike the full AC model, the DC approximation alters the physical representation of the system, but it captures the essential mechanism driving active power flows and congestion patterns in high-voltage networks.

When the DC load flow equations are combined with a cost-minimization problem for generators, we obtain the DC Optimal Power Flow (DC OPF). The DC OPF determines the cost-minimizing dispatch of generation subject to power balance and line flow limits. Because the model is linear, it can be solved efficiently even for large systems. Importantly, the dual variables associated with the nodal power balance constraints become nodal electricity prices, also known as locational marginal prices. These prices reflect both generation costs and congestion in the network. When a transmission constraint becomes binding, cheap generation may no longer be able to serve all demand, more expensive units must be dispatched elsewhere, and prices begin to differ across locations. The resulting price differences generate congestion rents.

An alternative way of representing network constraints is the Available Transfer Capacity (ATC) approach, where limits are imposed on transfers between predefined zones. While easier to implement, ATC does not fully respect the physical laws governing loop flows and may produce solutions that are not feasible in the real network. In contrast, DC OPF respects the linearized physical structure of the grid and provides a consistent link between dispatch, flows, and prices. This explains why nodal pricing systems use DC OPF and why even zonal systems increasingly rely on flow-based representations.

The key takeaway is that the DC load flow is not merely a technical simplification; it is the foundation that makes modern electricity market modeling possible. By linearizing the physical grid in a meaningful way, it allows economic optimization and network physics to be solved simultaneously. Understanding DC load flow and DC OPF is therefore essential for analyzing congestion, price formation, welfare effects, and the interaction between market design and physical system operation in contemporary power markets.

Introduction

In this Power Markets course, we frequently use the DC power flow and the Optimal Power Flow (OPF) framework to represent the transmission grid. These models form the backbone of modern electricity market analysis, as they allow us to link economic decisions, such as generation dispatch and pricing, to the physical constraints of the power system. While DC power flow has been briefly introduced in TET4205 Power System Analysis, that course is not a prerequisite here. Therefore, this note provides the necessary background to ensure that all students have a solid understanding of the concepts required for market applications.

We begin with a short overview of the AC load flow model to establish the physical foundations of power flow analysis. We then introduce the decoupled load flow and show how, under reasonable assumptions for high-voltage transmission systems, the AC equations can be simplified to obtain the DC load flow model. This progression is intended to clarify the assumptions behind the DC approximation and to explain why it is both reasonable and useful in a market context.

Although the derivation from AC to DC load flow is included to provide conceptual consistency and physical intuition, the main focus of this course is on the DC load flow and DC OPF formulations themselves. You are therefore expected to understand the assumptions underlying the DC model and to be comfortable working directly with its equations in market analyses.

The material presented here is largely based on [1], with adaptations and emphasis tailored to electricity market applications.

AC Load Flow

The calculations necessary to find the flows in a power grid are called "load flow calculations". The input data to these calculations consist of the network topology, the parameters necessary to build the so-called admittance matrix, and the node information. The admittance matrix describes the admittances between two nodes in the network (see below). The results of the calculations are node voltages, branch currents, branch powers and power losses. In a traditional load flow calculation, active generator power is given for all generators except the so-called slack bus, which makes up for the (unknown) losses in the network. In this case, the calculation results include the *reactive* power on all generators and the active power on the slack bus. In an *optimal* power flow calculation, however, the active power on all generators results from the calculations[1].

Figure 1 illustrates the input and results from the load flow calculations.

[1] In traditional power engineering, OPF was only used to determine the reactive power production on the generators, while the active power production was given. With the introduction of markets, determination of the active power production on the generators has become a more and more important area for OPF.

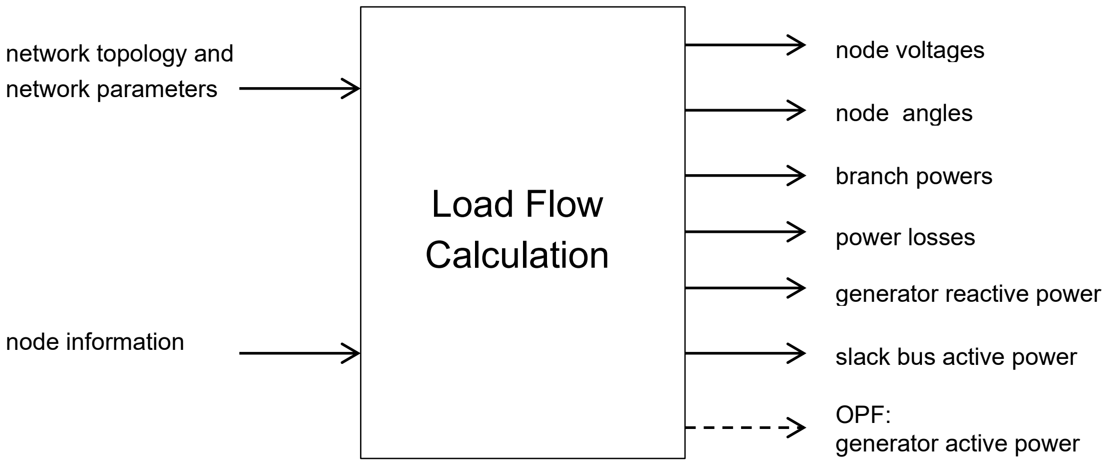

Figure 1: Load flow calculation: input data and results [1]

The admittance matrix describes the relationship between the currents I injected at the network nodes and V the node voltages[2]:

$$\mathbf{I} = \mathbf{YV} \leftrightarrow \quad \begin{bmatrix} I_1 \\ I_2 \\ . \\ . \\ I_N \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} & . & . & Y_{1N} \\ Y_{21} & Y_{22} & . & . & Y_{2N} \\ . & . & . & . & . \\ . & . & . & . & . \\ Y_{N1} & Y_{N2} & . & . & Y_{NN} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ . \\ . \\ V_N \end{bmatrix} \tag{1}$$

Here n is the total number of network nodes. The admittance matrix is built up as follows:

- The element on the main diagonal Y_{ii} equals the sum of all admittances directly connected to node i
- The off-diagonal element Y_{ij} equals the negative value of the net admittance connected between node i and node j (note that the off-diagonal element Yji has the same value).

The elements of the admittance matrix **Y** can be written as:

$$Y_{in} = |Y_{in}|\angle\theta_{in} = |Y_{in}|(\cos\theta_{in} + j\sin\theta_{in}) = G_{in} + jB_{in} \tag{2}$$

[2] To avoid numerical problems during the calculations, all values are normally in a so called *per unit* system during the actual computations. Reference for active and reactive power is normally 100 or 1000 MVA, depending on the size of the power system. Reference for the voltage is the nominal voltage at each voltage level in the system. Reference for impedances is then V^2/S, where V is the voltage reference and S the power reference. We do not go further into this in this course, but it is important to realize the use of per unit (pu), because actual values of impedances may or may not be given as pu.

where G_{in} is the conductance and B_{in} the susceptance. In general, we have:

$$Y = \frac{I}{V} = G + jB \tag{3}$$

and

$$Z = \frac{V}{I} = R + jX \tag{4}$$

which means that

$$G = \frac{R}{R^2 + X^2} \quad \text{and} \quad B = -\frac{X}{R^2 + X^2} \tag{5}$$

We see that for a purely resistive impedance $G = \frac{1}{R}$ and for a pure reactive impedance $B = -\frac{1}{X}$.
The voltages in the network nodes can be expressed as:

$$V_i = |V_i|\angle\delta_i = |V_i|(\cos\delta_i + j\sin\delta_i) \tag{6}$$

From equation (1) the injected current at node *I* can be obtained:

$$I_i = Y_{i1}V_1 + Y_{i2}V2 + ... + Y_{in}V_n = \sum_{n=1}^{N} Y_{iN}V_N \tag{7}$$

The complex power injected in node *I* is:

$$S_i = V_i I_i^* = P_i + jQ_i \tag{8}$$

Substituting the equations for the voltage at node *i* (equation (6)) and the injected current at node *i* (equation (7)) gives:

$$\begin{aligned} S_i &= |V_i|\angle\delta_i \sum_{n=1}^{N} Y_{in}^* V_j^* = \sum_{n=1}^{N} |V_i V_n Y_{in}| \angle(-\theta_{in} - \delta_n + \delta_i) \\ &= \sum_{n=1}^{N} |V_i V_n Y_{in}| \left[\cos(\theta_{in} - \delta_n + \delta_i) + j\sin(-\theta_{in} - \delta_n + \delta_i)\right] \\ &= \sum_{n=1}^{N} |V_i V_n Y_{in}| \left[\cos(\theta_{in} + \delta_n - \delta_i) - j\sin(\theta_{in} + \delta_n - \delta_i)\right] \end{aligned} \tag{9}$$

Thus, the expressions for active and reactive power are:

$$
\begin{aligned}
P_i &= \sum_{n=1}^{N} \left| V_i V_n Y_{in} \right| \cos(\theta_{in} + \delta_n - \delta_i) \\
&= \left| V_i \right|^2 G_{ii} + \sum_{\substack{n=1 \\ \neq i}}^{N} \left| V_i V_n Y_{in} \right| \cos(\theta_{in} + \delta_n - \delta_i)
\end{aligned} \tag{10}
$$

$$
\begin{aligned}
Q_i &= -\sum_{n=1}^{N} \left| V_i V_n Y_{in} \right| \sin(\theta_{in} + \delta_n - \delta_i) \\
&= -\left| V_i \right|^2 B_{ii} - \sum_{\substack{n=1 \\ \neq i}}^{N} \left| V_i V_n Y_{in} \right| \sin(\theta_{in} + \delta_n - \delta_i)
\end{aligned} \tag{11}
$$

These last two equations constitute the load flow equations. They represent the computed active and reactive power injections at node I, as a function of, in principle, all the complex node voltages in the network. These equations are solved in an iterative scheme, normally the so-called Newton-Raphson method. We will not go into the solving of this (AC) load flow equations but use them to derivate the DC power flow equations.

The load flow equations are generally solved by starting with some initial estimates of the unknown voltage angles δ_i and voltage magnitudes $|V|$ and calculating the resulting active and reactive power injections using the equations above. Subsequently, new values of the voltage angles and magnitudes are calculated using the derivatives of (10) and (11) and the relation

$$
\mathbf{J}\boldsymbol{\Delta}\mathbf{x} = \begin{bmatrix} \boldsymbol{\Delta}\mathbf{P}(\mathbf{x}) \\ \boldsymbol{\Delta}\mathbf{Q}(\mathbf{x}) \end{bmatrix} \tag{12}
$$

Where **J** is the Jacobian matrix of the derivatives of (10) and (11), $\boldsymbol{\Delta}\mathbf{x}$ the change in the voltage angles and magnitudes and ΔP and ΔQ the deviations from the desired values of P and Q[3]:

[3] In load flow programs it is common to divide the nodes in:

- the slack node with |V| and δ given and P and Q unknown
- generator or "PV" nodes, with P and |V| given and Q and δ unknown
- load or "PQ" nodes, with P and Q given and |V| and δ unknown

In that case the δ-part of **x** exists of the generator nodes only and the |V|-part of the load nodes. Here we include all nodes as a preparation to the transition to the DC power flow. The scheme as presented here can also be solved directly, but leads to a lot of unnecessary multiplications with zero.

$$\mathbf{x} = \begin{bmatrix} \boldsymbol{\delta} \\ |\mathbf{V}| \end{bmatrix} = \begin{bmatrix} \delta_2 \\ . \\ \delta_N \\ |V|_2 \\ . \\ |V|_N \end{bmatrix} \tag{13}$$

Note that the numbering of the **x** vector starts at 2, because the voltage angle and magnitude at the reference (slack) bus are given.

The Jacobian matrix **J** can be written as:

$$\mathbf{J} = \begin{bmatrix} \mathbf{J_{11}} & \mathbf{J_{12}} \\ \mathbf{J_{21}} & \mathbf{J_{22}} \end{bmatrix} \tag{14}$$

where $\mathbf{J_{11}}$ exists of the derivatives of the active power with respect to the angles, $\mathbf{J_{12}}$ the derivatives of the active power with respect to the voltage magnitudes, $\mathbf{J_{21}}$ the derivatives of the reactive power with respect to the angles and $\mathbf{J_{22}}$ the derivatives of the reactive power with respect to the voltage magnitudes. On the basis of (10) and (11) we can find the following derivatives:

$$\frac{\partial P_i}{\partial \delta_i} = \sum_{\substack{n=1 \\ \neq i}}^{N} |V_i V_n Y_{in}| \sin(\theta_{in} + \delta_n - \delta_i) \tag{15}$$

$$\frac{\partial P_i}{\partial \delta_j} = -|V_i V_j Y_{ij}| \sin(\theta_{ij} + \delta_j - \delta_i) \tag{16}$$

$$\frac{\partial Q_i}{\partial \delta_i} = \sum_{\substack{n=1 \\ \neq i}}^{N} |V_i V_n Y_{in}| \cos(\theta_{in} + \delta_n - \delta_i) \tag{17}$$

$$\frac{\partial Q_i}{\partial \delta_j} = -|V_i V_j Y_{ij}| \cos(\theta_{ij} + \delta_j - \delta_i) \tag{18}$$

$$\frac{\partial P_i}{\partial |V_i|} = 2|V_i| G_{ii} + \sum_{\substack{n=1 \\ \neq i}}^{N} |V_n Y_{in}| \cos(\theta_{in} + \delta_n - \delta_i) \tag{19}$$

$$\frac{\partial P_i}{\partial |V_j|} = |V_i Y_{ij}| \cos(\theta_{ij} + \delta_j - \delta_i) \tag{20}$$

$$\frac{\partial Q_i}{\partial |V_i|} = -2|V_i| B_{ii} - \sum_{\substack{n=1 \\ \neq i}}^{N} |V_n Y_{in}| \sin(\theta_{in} + \delta_n - \delta_i) \tag{21}$$

$$\frac{\partial Q_i}{\partial |V_j|} = -|V_i Y_{ij}| \sin(\theta_{ij} + \delta_j - \delta_i) \tag{22}$$

So $\mathbf{J_{11}}$ exists of elements of type (15) and (16), $\mathbf{J_{12}}$ exists of elements of type (19) and (20), $\mathbf{J_{21}}$ of elements of type (17) and (18) and finally, $\mathbf{J_{22}}$ exists of elements of type (21) and (22).

Examples can be found in, e.g. [1], [2] and many other references.

Decoupled load flow

AC load flow, as described above, is a relatively time-consuming algorithm because the Jacobi matrix must be recalculated for each iteration due to its dependence on the state variables δ and $|V|$. This can be avoided by taking into account that there is a kind of "decoupling" between the active power and the voltage angles on the one hand and the reactive power and voltage magnitudes on the other when specific properties of power systems are taken into account:

- The resistance of the overhead transmission lines is much smaller than its reactance
- The differences between the voltage angles are typically small

As a result, the elements of the off-diagonal sub-matrices $\mathbf{J_{12}}$ and $\mathbf{J_{21}}$ are typically an order of magnitude less than the diagonal elements of the $\mathbf{J_{11}}$ and $\mathbf{J_{22}}$, and in the "decoupled" load flow, these elements are set equal to zero. The equation (12) results in the following two equations, using (14):

$$\begin{aligned} \mathbf{J}_{11}\,\boldsymbol{\Delta\delta} &= \boldsymbol{\Delta}\mathbf{P} \\ \mathbf{J}_{22}\,\boldsymbol{\Delta}|\mathbf{V}| &= \boldsymbol{\Delta}\mathbf{Q} \end{aligned} \tag{23}$$

This means that the active power mismatches are used to calculate the voltage-angle corrections, whereas the reactive power mismatches are applied only to calculate the voltage-magnitude corrections. However, the elements of the matrices $\mathbf{J_{11}}$ and $\mathbf{J_{22}}$ still depend on both the voltage magnitude and the voltage angle, meaning that they have to be recalculated in each iteration of the load flow algorithm. In order to obtain an efficient computational procedure, further simplifications can be made in these matrices when we take into account that during normal system operation:

- The line susceptances B of Y_{bus} are much larger than the line conductances G of Y_{bus}
- The differences between the voltage angles are small
- The reactive power injected into a node is much smaller than the reactive flow that would result if all lines connected to that bus were short-circuited to reference

As a result of the first two points $G_{ij}\sin(\delta_j - \delta_i) \ll B_{ij}\cos(\delta_j - \delta_i)$ and $\sin(\delta_j - \delta_i) \approx \delta_j - \delta_i$ and $\cos(\delta_j - \delta_i) \approx 1$. The last point means $Q_i \ll |V_i|^2 B_{ii}$. Using this, the elements in $\mathbf{J_{11}}$ can be simplified as follows:

$$\begin{aligned}\frac{\partial P_i}{\partial \delta_i} &= \sum_{\substack{n=1\\ \neq i}}^{N} \left|V_i V_n Y_{in}\right| \sin(\theta_{in} + \delta_n - \delta_i) \\ &= -Q - \left|V_i\right|^2 B_{ii} \approx -\left|V_i\right|^2 B_{ii}\end{aligned} \tag{24}$$

$$\begin{aligned}\frac{\partial P_i}{\partial \delta_j} &= -\left|V_i V_j Y_{ij}\right| \sin(\theta_{ij} + \delta_j - \delta_i) \\ &= -\left|V_i V_j Y_{ij}\right| \left[\sin(\theta_{ij})\cos(\delta_j - \delta_i) + \cos(\theta_{ij})\sin(\delta_j - \delta_i)\right] \\ &= -\left|V_i V_j\right| \left[B_{ij}\cos(\delta_j - \delta_i) + G_{ij}\sin(\delta_j - \delta_i)\right] \approx -\left|V_i V_j\right| B_{ij}\end{aligned} \tag{25}$$

To find simplified expressions of the elements in $\mathbf{J}_{22}$, it is convenient to multiply the partial derivatives with the voltage magnitude:

$$\begin{aligned}\left|V_i\right| \frac{\partial Q_i}{\partial \left|V_i\right|} &= -2\left|V_i\right|^2 B_{ii} - \sum_{\substack{n=1\\ \neq i}}^{N} \left|V_i V_n Y_{in}\right| \sin(\theta_{in} + \delta_n - \delta_i) \\ &= Q_i - \left|V_i\right|^2 B_{ii} \approx -\left|V_i\right|^2 B_{ii}\end{aligned} \tag{26}$$

$$\begin{aligned}\left|V_j\right| \frac{\partial Q_i}{\partial \left|V_j\right|} &= -\left|V_i V_j Y_{ij}\right| \sin(\theta_{ij} + \delta_j - \delta_i) \\ &= -\left|V_i V_j Y_{ij}\right| \left[\sin(\theta_{ij})\cos(\delta_j - \delta_i) + \cos(\theta_{ij})\sin(\delta_j - \delta_i)\right] \\ &= -\left|V_i V_j\right| \left[B_{ij}\cos(\delta_j - \delta_i) + G_{ij}\sin(\delta_j - \delta_i)\right] \approx -\left|V_i V_j\right| B_{ij}\end{aligned} \tag{27}$$

Using this, and filling in $\mathbf{J}_{11}$ and $\mathbf{J}_{22}$ explicitly in (23), we get:

$$\begin{bmatrix} -\left|V_2\right|^2 B_{22} & . & . & -\left|V_2 V_N\right| B_{2N} \\ . & . & . & . \\ . & . & . & . \\ -\left|V_N V_2\right| B_{N2} & . & . & -\left|V_N\right|^2 B_{NN} \end{bmatrix} \begin{bmatrix} \Delta\delta_2 \\ . \\ . \\ \Delta\delta_N \end{bmatrix} = \begin{bmatrix} \Delta P_2 \\ . \\ . \\ \Delta P_N \end{bmatrix} \tag{28}$$

$$\begin{bmatrix} -\left|V_2\right| B_{22} & . & . & -\left|V_N\right| B_{2N} \\ . & . & . & . \\ . & . & . & . \\ -\left|V_N\right| B_{N2} & . & . & -\left|V_N\right| B_{NN} \end{bmatrix} \begin{bmatrix} \Delta\left|V_2\right| \\ . \\ . \\ \Delta\left|V_N\right| \end{bmatrix} = \begin{bmatrix} \Delta Q_2 \\ . \\ . \\ \Delta Q_N \end{bmatrix} \tag{29}$$

When we divide each row in this system of equations by the voltage magnitude that is common in this row and makes the remaining voltage magnitudes in the left-hand side of equation (28) equal to 1.0 per unit, we obtain a decoupled system of equations:

$$\begin{bmatrix} -B_{22} & . & . & -B_{2N} \\ . & . & . & . \\ . & . & . & . \\ -B_{N2} & . & . & -B_{NN} \end{bmatrix} \begin{bmatrix} \Delta\delta_2 \\ . \\ . \\ \Delta\delta_N \end{bmatrix} = \begin{bmatrix} \Delta P_2/|V_2| \\ . \\ . \\ \Delta P_N/|V_N| \end{bmatrix} \quad (30)$$

,

$$\begin{bmatrix} -B_{22} & . & . & -B_{2N} \\ . & . & . & . \\ . & . & . & . \\ -B_{N2} & . & . & -B_{NN} \end{bmatrix} \begin{bmatrix} \Delta|V_2| \\ . \\ . \\ \Delta|V_N| \end{bmatrix} = \begin{bmatrix} \Delta Q_2/\Delta|V_2| \\ . \\ . \\ \Delta Q_N/\Delta|V_N| \end{bmatrix} \quad (31)$$

Both matrices are constant and depend on the grid parameters only. This means that the matrices have to be calculated only once, and this results in a faster load flow algorithm, despite the fact that more iterations may be needed to find the solution.

It is important to realize that the approximations are made *only in the Jacobian matrix*, while the power mismatches are evaluated without approximation. Therefore, only the number of iterations is affected, but not the final result. So the (fast) decoupled load flow calculation still results in the exact AC load flow result but uses a different and normally faster algorithm.

Before introducing the DC load flow formally, it is useful to reflect on what we are trying to achieve. In electricity markets, we are primarily interested in active power flows, since active power determines production costs, prices, congestion, and welfare. While the full AC load flow model gives a very accurate description of the physical system, it is nonlinear and computationally demanding. In market simulations and market clearing, we often need to solve optimization problems repeatedly and for large systems. Therefore, we look for a simplified representation that captures the essential physics relevant for active power dispatch, while remaining computationally tractable.

Physically, active power flows in transmission systems are mainly driven by voltage angle differences rather than voltage magnitude differences. Moreover, in high-voltage transmission grids, line resistances are typically small compared to reactances, and voltage magnitudes are tightly controlled around 1 per unit. Under these normal operating conditions, the nonlinear AC power flow equations can be simplified considerably. The DC load flow builds on these observations: it preserves the key relationship between power injections and voltage angles while linearizing the model. This linear structure is what makes the DC load flow particularly suitable for integration with optimization models such as DC Optimal Power Flow (DC OPF).

> The full AC load flow model gives an accurate description of voltage magnitudes, voltage angles, active power, and reactive power. However, it is nonlinear and computationally intensive.
> In electricity market applications, we are primarily interested in active power flows, congestion, and prices. This motivates the search for a simplified model that preserves the essential physics relevant for active power dispatch while remaining computationally tractable.

DC load flow

In situations where a lot of load flow computations have to be made, as, e.g. for reliability computations or security analyses, a linear approximation of the load flow problem can be made to save computation time: this is called the DC load flow because of its similarity to calculations in a DC network. So although the name may suggest differently, the DC load flow is not for DC networks (as in Section 7.3.8 in [4], which uses a resistive DC network).

An important advantage of the DC load flow is that it can be combined with other linear optimizations problems like optimal dispatch.

The DC load flow is principally different from the decoupled load flow. In the DC load flow, the non-linear load flow equations are linearized to ease the calculation and to speed up the computation. This means that the actual model of the power system is altered, and this affects the solution. In the decoupled load flow, the non-linear equations are solved iteratively, and the model of the power system remains unchanged.

It is possible to extend the DC load flow with calculations of the reactive power and voltage magnitudes, cf. [1], Section 6.2.4, but here we will only focus on active power and voltage angles, an approach that is frequently used in market simulations and even market clearing.

In the DC load flow method, the following approximations are made:

- All node voltage magnitudes are set to 1 pu
- The resistances of the transmission lines are neglected: $Y_{ij} = G_{ij} + jB_{ij} = |Y_{ij}| \angle \theta_{ij}$ with $G_{ij} = 0$ and $\theta_{ij} = \pi/2$ rad
- The differences between the voltage angles are small: $\sin(\delta_j - \delta_i) \approx (\delta_j - \delta_i)$ and $\cos(\delta_j - \delta_i) \approx 1$

Under these assumptions, the active power load flow equation (10) can be written as:

$$
\begin{aligned}
P_i &= \sum_{n=1}^{N} \left|V_i V_n Y_{in}\right| \cos(\theta_{in} + \delta_n - \delta_i) \\
&\approx \sum_{\substack{n=1 \\ \neq i}}^{N} \left|Y_{in}\right| \cos(\frac{\pi}{2} + \delta_n - \delta_i) = \sum_{\substack{n=1 \\ \neq i}}^{N} \left|Y_{in}\right| \sin(-\delta_n + \delta_i) \\
&\approx \sum_{\substack{n=1 \\ \neq i}}^{N} \left|Y_{in}\right| (\delta_i - \delta_n)
\end{aligned} \tag{32}
$$

This is a linear relationship between the active power injections and the busbar voltage angles. The flow on an interconnection is given by:

$$
P_{ij} = -b_{ij}\left(\delta_i - \delta_j\right) = \frac{1}{x_{ij}}\left(\delta_i - \delta_j\right) \tag{33}
$$

using (5) and R = 0.

The DC load flow is a linear approximation of the AC model.

It assumes:

- Voltage magnitudes are fixed at 1 pu
- Line resistances are negligible
- Angle differences are small

Under these assumptions, active power injections become linearly related to voltage angles.

This linearity makes the DC model compatible with optimization problems and therefore highly suitable for electricity market analysis.

DC Optimal Power Flow

We are now in a position to describe the DC Optimal Power Flow (DC OPF). Because it is linear, the DC power flow can easily be combined with, e.g. the minimization of the production (or offer) costs in a power market.

Assume a power system with *M* generators with linear costs, i.e. a constant marginal cost. The minimum cost to run the system with a given load, without taking into account the network, results from the following optimal dispatch problem:

$$
\begin{aligned}
& MIN\ f(\mathbf{P}) \\
& subject\,to: \quad \sum \mathbf{P} = \mathbf{D} \\
& \qquad\qquad\qquad \mathbf{P_{min}} \le \mathbf{P} \le \mathbf{P_{max}}
\end{aligned} \tag{34}
$$

Here f is the cost function of all generators, P_i their actual generation, D the system demand and P_{min} and P_{max} the minimum and maximum production of the generators, respectively. This very simplified formulation assumes that either P_{min} is set equal to zero for all generator (possibly resulting in some generators running between zero and their real minimum power), or that the actual generators that are running have been pre-determined in a unit commitment procedure. Here we disregard these complications, and remind of the result that in the solution, generators either will have the same marginal cost, or run at P_{max} with a marginal cost below the "system" marginal cost or run at P_{min} (possibly zero) with a marginal cost above the "system" marginal cost.

The easiest way to include grid constraints in this problem is to add the maximum flows as a constraint to the problem formulation. In fact, this is the approach used in the Nord Pool system (apart from the fact that there are a number of additional complications related to the bids and offers; these are however not essential for the principal treatment of the grid constraints). We can formulate this problem by defining a bus-incidence matrix **I** that defines the network topology. The dimension of this matrix is $N \times L$, where L is the number of lines in the system, and N the number of nodes. An element in **I** is 1 if it represents the *from* node of a line and -1 if it represents the *to* node of a line. The line flows P_{ij} are given in the vector **FL**. The optimal dispatch with network constraints can then be formulated as:

$$
\begin{aligned}
& MIN\ f(\mathbf{P}) \\
& subject\,to: \quad \mathbf{P} - \mathbf{D} = \mathbf{I} \times \mathbf{FL} \\
& \qquad\qquad\qquad \mathbf{FL_{min}} \le \mathbf{FL} \le \mathbf{FL_{max}} \\
& \qquad\qquad\qquad \mathbf{P_{min}} \le \mathbf{P} \le \mathbf{P_{max}}
\end{aligned} \tag{35}
$$

The first constraint just says that the difference between production and demand in a node equals the flow out from that node, and this flow is restricted between its minimum and maximum value. Normally $FL_{min} = - FL_{max}$. Note that this is *not* (yet) the DC OPF problem, it is just the optimal dispatch problem with grid constraints included. It is sometimes referred to as the "ATC approach", where ATC stands for Available Transfer Capacity. The advantage of this method is that it is easy to understand, once you accept the transfer capacities as given. The *disadvantage* however, is that the solution does not follow the physical reality defined by Kirchhoff's voltage and current laws. This means that a market cleared according to this procedure may result in a solution that is not physically possible and has to be adjusted. Another problem with the method is that it is often difficult to set the transfer limits in such a way that they reflect the physical reality, especially in heavily mashed networks.

A better approach is therefore to use the DC OPF, which uses the DC load flow as described above together with the optimal dispatch problem. In principle, this means including equations (32) and (33) in the problem description, resulting in the following problem:

$$
\begin{aligned}
&MIN\ f(\mathbf{P}) \\
&subject\,to: \quad \mathbf{P} - \mathbf{D} = j\mathbf{Y} \times \mathbf{\Delta} \\
&\qquad\qquad\quad \mathbf{FL_{MIN}} \leq \mathbf{F\Delta} \leq \mathbf{FL_{MIN}} \\
&\qquad\qquad\quad \mathbf{P_{min}} \leq \mathbf{P} \leq \mathbf{P_{max}}
\end{aligned}
\tag{36}
$$

Here $\mathbf{\Delta}$ is the vector of the voltage angles. The elements of the $L \times N$ matrix $\mathbf{F}$(connection matrix) are equal to $|b_{ij}|$ if i is the *from* node of line l, and $-|b_{ij}|$ if i is the *to* node of the line. So $\mathbf{F}$ has the same structure as the transposed of the matrix $\mathbf{I}$ above, but instead of +1 and -1 it contains the line susceptances.

The ATC approach limits transfers between areas but does not fully reflect the physical laws governing loop flows in a meshed network.

DC OPF, in contrast, incorporates a physical (linearized) representation of the grid and respects both Kirchhoff's current and voltage laws.

As a result:

- ATC solutions may be economically optimal but physically infeasible.
- DC OPF solutions are physically consistent within the DC approximation.

Examples

We will now illustrate the various concepts with a small example grid. We will use the same example grid as in [3]:

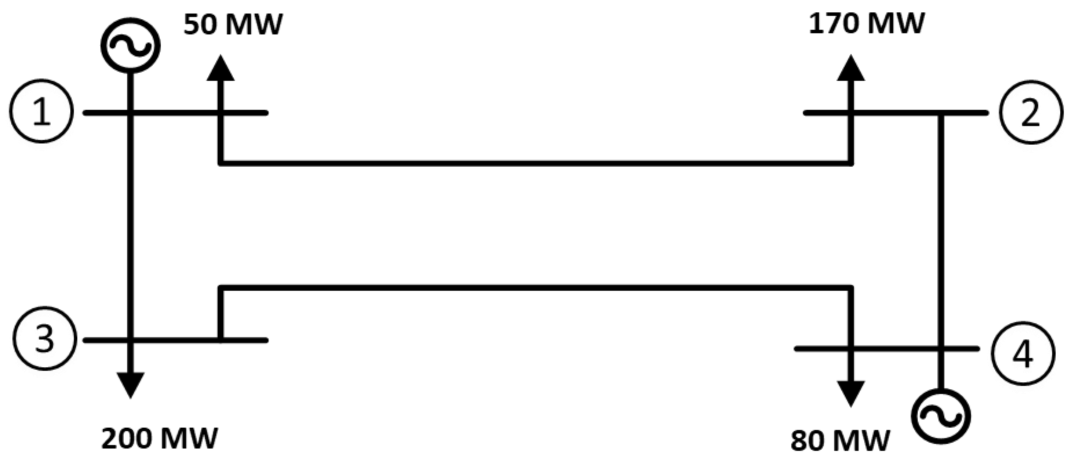

Figure 2: Example grid [3]

The susceptances b_{ij} are given by the inverse of the reactances x_{ij}, ignoring the resistances:

line	from-to	B (per unit)
1	1-2	-19.84
2	1-3	-26.88
3	2-4	-26.88
4	3-4	-15.72

Instead of a quadratic cost function for the generators we will assume a constant marginal cost of 50 €/MWh for generator 1 and 45 €/MWh for 4. The per-unit base is 100 MVA for power and 230 kV for voltage.

Least cost solution without taking into account the transmission network

This is a trivial problem; without constraints on the generator capacities of course, all power will be delivered by G4. Without losses, the production is 500 MW, and the cost 500 x 45 = 22500 €/hr. There is one uniform price in the system, equal to the marginal cost of G4, 45 €MWh.

The actual grid flow, using DC Power Flow is given below:

From Node	To Node	Flow (MW)
1	2	-60.7
1	3	10.7
2	4	-230.7
3	4	-189.3

Table 1: Example solution without grid consideration

Least cost solution with transmission constraints

We assume that the maximum flow on lines 1 and 4 is equal to 100 MW.
This case corresponds to equation (35). The matrix $\mathbf{I}$ must first be determined:

$$\mathbf{I} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & -1 \end{bmatrix} \tag{37}$$

Using this, we get the following problem:

$$\begin{aligned} &MIN\ c_1P_1 + c_4P_4 \\ &subject\,to: \quad && P_1 - D_1 = P_{12} + P_{13} \\ &&& P_2 - D_2 = -P_{12} + P_{24} \\ &&& P_3 - D_3 = -P_{13} + P_{34} \\ &&& P_4 - D_4 = -P_{24} - P_{34} \\ &&& -100 \le FL_{12} \le +100 \\ &&& -100 \le FL_{34} \le +100 \end{aligned} \tag{38}$$

Solving this problem[4] yields the following solution:

[4] The problem is solved using an LP solver. Although it is possible to solve this small example by hand, we do not go into the details of that.

Node	Generation (MW)	Demand (MW)	Price (€/MWh)
1	50	50	50
2	0	170	45
3	0	200	50
4	450	80	45

From Node	To Node	Flow (MW)	Maxflow (MW)	Dual (€/MWh)
1	2	-100	100	5
1	3	100		
2	4	-270		
3	4	-100	100	5

Table 2: Example solution with ATC approach

The problem now becomes more interesting. Because of the transmission constraints, it is not possible to produce all power on G4, which now produces 450 MW, while G1 produces 50 MW. There are now two prices in the system: nodes 2 and 4 have the lower price of 45 €/MWh corresponding to the marginal cost of G4, and nodes 1 and 3 have the higher price of 50 €/MWh, corresponding to the marginal cost of G1. The dual values of the line constraints correspond to the difference between these two prices.

The total cost is now 450 x 45 + 50 x 50 = 22750 €/hr. Thus the (societal) cost of the constraint is 250 €/hr. However, the cost to consumers is 250 x 5 = 1250 €/hr. This is equal to the cost increase of 250 plus the congestion rent of 1000, cf. Section 7.2.2 in [4].

Least cost solution with DC OPF

We now solve the problem with transmission constraints as above using DC OPF, corresponding to equation (36). The **Y** and **F** matrices are as follows (cf. [3]):

$$\mathbf{Y} = j\begin{bmatrix} -46.72 & 0 & 26.88 \\ 0 & -42.60 & 15.72 \\ 26.88 & 15.72 & -42.60 \end{bmatrix} \text{ and } \mathbf{F} = \begin{bmatrix} -19.84 & 0 & 0 \\ 0 & -26.88 & 0 \\ 26.88 & 0 & -26.88 \\ 0 & 15.72 & -15.72 \end{bmatrix}$$

Note that **Y** is given as a 3×3 matrix instead of a 4×4 matrix because the first line and row have been removed due to the fact that bus 1 is chosen as the reference bus. For the same reason, the first column of **F** has been removed. Because of this, the sum of the values in the rows of both matrices is not always zero, as it would have been in the complete 4×4 matrices. The values are slightly different from those given in [3] because we have chosen to use the susceptances instead of the admittances.

The problem formulation becomes:

$$
\begin{aligned}
& MIN\ c_1P_1 + c_4P_4 \\
& subject\,to: \quad P_1 = D_1 + \sum_{i=2}^{4}(D_i - P_i) \\
& \qquad\qquad\quad P_2 - D_2 = 46.72\delta_2 - 26.88\delta_4 \\
& \qquad\qquad\quad P_3 - D_3 = 42.60\delta_3 - 15.72\delta_4 \\
& \qquad\qquad\quad P_4 - D_4 = -26.88\delta_2 - 15.72\delta_3 + 42.60\delta_4 \\
& \qquad\qquad\quad -100 \le 19.84\delta_2 \le +100 \\
& \qquad\qquad\quad -100 \le 15.72\delta_3 - 15.72\delta_4 \le +100
\end{aligned}
\tag{39}
$$

Solving this, we get the following solution:

Node	Generation (MW)	Demand (MW)	Angle (Radians)	Angle (Degrees)	Price (€/MWh)
1	192	50	0	0	50
2	0	170	-0.0212	-0.61	47.1
3	0	200	-0.0372	-1.07	52.1
4	308	80	0.0264	0.76	45

From Node	To Node	Flow (MW)	Maxflow (MW)	Dual (€/MWh)
1	2	42	100	0
1	3	100		
2	4	-128		
3	4	-100	100	10.8

Table 3: Example solution with DC OPF

The total cost is now 23460.2 €/hr.

It can easily be verified that the solution in

Table 3 satisfies equations (39). In this very simple example, it would actually be possible to calculate the solution by setting up the Lagrangian and solving the resulting equations.

The solution shows some interesting issues:

- It is not possible to obtain the previous solution in the real network because the flow pattern resulting from the transport constraints does not conform to the load flow equations.

- Because of this, the real solution becomes much more expensive than the solution given by the ATC procedure.
- Prices are different in all nodes – the price in node 3 is even higher than the highest marginal generation cost, which is caused by the constraints imposed by the load flow equations.

In general, one constraint in a grid is in principle enough to cause different prices in all nodes with the DC OPF solution because all nodes are "coupled" through the load flow equations.

Of course, the TSOs are aware of the discrepancy between the ATC solution and the real network solutions, and they try to avoid such situations by adjusting the ATC values to what is possible in reality. Assume that the TSO anticipates the situation, and reduces the market capacity on the interconnection between node 1 and 2. The obvious choice would be 42 MW, but if that is used, it turns out that the production on G1 is 108 MW, and the flow becomes 42 MW from node 2 to node 1. So the best the TSO can do is to constrain the value to zero, giving the following result:

Node	Generation (MW)	Demand (MW)	Price (€/MWh)
1	150	50	50
2	0	170	45
3	0	200	50
4	350	80	45

From Node	To Node	Flow (MW)	Maxflow (MW)	Dual (€/MWh)
1	2	0	0	5
1	3	100		
2	4	-170		
3	4	-100	100	5

Table 4: Example solution with ATC approach and additional grid constraint

This is much closer to the physically possible solution, but still this solution is not realizable in the real network. The same generation dispatch can be obtained by setting the constraint on line 3-4 to 58 MW:

Node	Generation (MW)	Demand (MW)	Price (€/MWh)
1	192	50	50
2	0	170	45
3	0	200	50
4	308	80	45

From Node	To Node	Flow (MW)	Maxflow (MW)	Dual (€/MWh)
1	2	0	0	5
1	3	142		
2	4	-170		
3	4	-58	58	5

Table 5: Example solution with ATC approach and sufficient grid constraints to support DC OPF solution

Assume now that this system uses area pricing (Section 7.2.2 in [4]), and that the area 1 consists of nodes 1 and 3 and area 2 of nodes 2 and 4. The areas will have a price of 50 and 45 €/MWh respectively. However, in the real solution as given by

Table 3, the flow on the line from node 1 to node 2 will go from high price to low price! This paradoxical result comes from the fact that the ATC approach is not able to take into account the real grid constraints. This often happens in reality, e.g. between areas NO1 and NO3 in Norway. Of course the total flow between the high price and low price areas is from low price to high price (the sum of the flows on lines 1-2 and 3-4), but on individual lines the flow can be against the price direction.

Another issue is that the TSO will not be able to exactly anticipate the future loads, and moreover will want to use ATC values that are stable for (at least) several hours (on most interconnections they are constant for large parts of the year). It might also be necessary to set limits on lines within areas to obtain the desired solution, while that is not possible in an area pricing model.

Consequently, the real solution obtained in the market may not be possible in reality, and the TSO must use a form for redispatch or buy-back to get a feasible solution, cf. [4], Section 7.2.3. If e.g. the solution in

Table 2 is the final result, the TSO would pay G1 50 €/MWh to increase production from 50 to 192 MW, and get paid 45 €/MWh from G4 to reduce production from 450 to 308 MW.

In DC OPF, the nodal prices emerge as shadow prices of the power balance constraints.
When transmission constraints bind:

- Prices differ across nodes
- Congestion rents arise
- Some generators may produce below their maximum even if they are cheapest

Even a single binding line constraint can influence prices at all nodes due to network coupling.
Thus, congestion fundamentally shapes price formation in electricity markets.

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